Spectral–Spatial MLP-Like Network With Reciprocal Points Learning for Open-Set Hyperspectral Image Classification

Yifan Sun, Bing Liu, Ruirui Wang, Pengqiang Zhang, and Mofan Dai

***Abstract*— In recent years, deep-learning-based hyperspectral image (HSI) classification methods have achieved significant development and gradually become widely applied. The existing advanced methods can achieve near-saturation performance with sufficient labels in a closed-set environment (CSE); i.e., training set and test set are all known categories of ground objects. However, the real world is usually open because of the diversity of land covers; i.e., test set exists unknown categories that are not labeled in the training set. Therefore, the prevalent advanced CSE methods still cannot effectively and robustly handle unknown categories of ground objects in an open-set environment (OSE). Therefore, we propose a spectral–spatial multiple layer perceptron (MLP)-like network with reciprocal points learning (SSMLP-RPL) to improve the performance of open-set HSI classification. First, a feature learning framework based on RPL is constructed to model the extra-category space and reduce the risk of open space. The learned feature space enables to enlarge the distance between the known and unknown categories. Besides, we further propose to utilize a learnable dynamic threshold of each known category to effectively distinguish the unknown categories and improve open performance of the model. Second, to enhance the capacity of feature learning, an SSMLP-like network is designed to capture the spectral–spatial feature merely with a series of fully connected (FC) layers, which mainly involve SpeFC and SpaFC two modules. Among them, the SpaFC module enables to model spacial semantics, and the SpeFC module enables to model long-distance spectral dependence. Extensive experiments on three benchmark HSIs show that SSMLP-RPL has a competitive performance both in CSE and OSE and even surpasses currently advanced closed-set and open-set HSI classification methods. As an end-to-end HSI classification framework of MLP backbone, SSMLP network can compete with the advanced works based on CNN and Transformer. The code will be open at https://github.com/sssssyf/SSMLP-RPL.**

***Index Terms*— Hyperspectral image (HSI), multiple layer perceptron (MLP)-like, open-set classification, reciprocal points learning (RPL), spectral–spatial.**

I. Introduction

CONTINUOUS development of Earth observation technology provides rich multimodal observation data for human cognition of complex surface scenes [1]. Hyperspectral imaging, as an essential source of multimodal remote sensing data, can capture the fine spectral features of ground objects [2]. It enables to recognize the surface materials, particularly those that have extremely similar spectral signatures in visual cues [red, green, blue (RGB)] [3]. Also, the capacity of hyperspectral image (HSI) processing and analysis will directly determine the level of ground object recognition [4]. The main branches in HSI processing involve denoising [5], dimensionality reduction [5], band selection [6], spectral unmixing [7], super-resolution fusion [8], and feature extraction [9]. Among them, HSI classification, as the last step to realize the recognition of ground object attributes, is a significant topic in HSI processing and analysis, which involves labeling the categories of ground objects pixel-by-pixel.

Due to the high-dimensional and nonlinear characteristics of data, feature extraction that focuses on a transformation of original information is pivotal for HSI classification. Early traditional feature extraction methods generally take the handcrafted feature design as guidelines or involve the statistic learning [10].

Initially, the spectrum of each HSI pixel (i.e., a 1-D vector of values in different spectral channels) is utilized as the processing unit for spectral feature extraction, such as principal component analysis (PCA) [11], linear discriminant analysis (LDA) [12], and so on. They have an apparent defect, neglecting the spatial dependencies between adjacent pixels. Thus, the pivotal problem of spatial feature extraction is how to properly model spatial contextual information rather than dimensionality reduction, such as shape and structure feature extraction methods represented by extended multiple profiles (EMPs) [13], texture feature extraction methods represented by Gabor [14], and local binary pattern (LBP) [15]. Furthermore, the joint extraction of spectral and spatial feature is extensively explored, such as 3-D LBP [16] and simultaneous spectral–spatial feature selection extraction (S3FSE) [17]. Although the early works mentioned above make a significant contribution to the field of hyperspectral, such handcrafted features heavily rely on the expertise. Besides, they are usually complex but shallow, which are difficult to adapt to a variety of hyperspectral scenarios.

Because of the powerful adaptive feature extraction ability of deep learning, data-driven HSI classification methods can automatically capture intrinsic features conducive to

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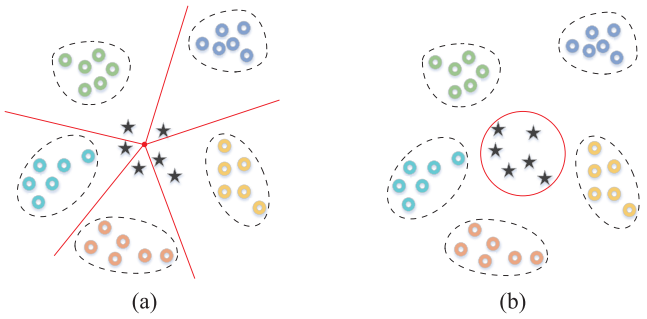


Fig. 1. Representation of closed-set and open-set classification. (a) Closed-set classification. (b) Open-set classification (the circles denote the known categories, and the stars denote the unknown categories).

identifying land covers [18]. In recent years, the continuous optimization of advanced network architecture and learning paradigms significantly have improved the classification accuracy of the model on benchmark HSI datasets.

Specifically, the early models usually take spectral sequences as processing units to capture exclusive spectral feature for land-cover classification [19], [20]. While limited by the intercategory spectrum similarity and the intracategory spectrum discrepancy, modeling the spatial information to improve the accuracy is a successful exploration in practice [21]. The convolutional neural network (CNN) that particularly specializes in capturing local spatial information is crucial in this process, and numerous advanced methods based on the CNN are constantly proposed to refresh accuracy records [22], [23], [24], [25], [26], [27], [28], [29]. Besides, methods based on other novel backbones are exploited constantly to bridge the limitation of CNN, such as graph convolutional network [30], [31], capsule network [32], long short-term memory (LSTM) [33], and Transformer [34], [35], [36], [37], [38]. Besides, the various attention mechanism (AM) modules are embedded in advanced networks to focus on hot areas and enhance the feature representation [39], [40]. In a closed-set environment (CSE), the considerable development of works above enables models to achieve near-saturation performance on benchmark HSI datasets with sufficient labels. Aiming at the condition of small samples, several advanced learning strategies are utilized to mitigate the ubiquitous dilemma of labeled samples scarcity, such as unsupervised learning [41], [42], self-supervised learning [43], [44], [45], semisupervised learning [46], meta-learning [47], [48], few-shot learning [49], [50], [51], transfer learning [52], [53], self-ensembling learning [54], and so on.

However, a common characteristic of currently prevalent works is the assumption of the CSE; i.e., training set and test set are all known categories of ground objects. This is namely the closed-set HSI classification task. Although the closed-set methods can usually achieve satisfactory accuracy under ideal conditions, the real world is typically open because of the diversity of land covers (several small ground objects may be easily ignored when manually labeling); i.e., the test set exists unknown categories that are not labeled in the training set. As shown in Fig. 1(a), because the typical closed-set deep-learning-based methods employed a linear classification layer or softmax function usually produce a maximal probability over the known categories for the unknown categories, the inevitably existing risk of open space will exacerbate the misidentification. Therefore, the prevalent advanced closed-set methods still cannot effectively and robustly handle the unknown categories of ground objects in open-set environment (OSE). As shown in Fig. 1(b), a robust HSI classification method should not only classify known categories correctly, but also identify the unknown categories for open-set HSI classification. Obviously, open-set HSI classification has more practical value than idealized closed-set classification, but it has received little attention in the past few years.

A task involves identifying the unknown categories and simultaneously classifying all test samples is defined as open set recognition (OSR) for the first time in [55]. Few works on such valuable but neglected task can be roughly divided into discriminative methods and generative methods. Discriminative methods usually construct a specific discriminative criterion to distinguish the known and unknown categories, such as the distribution of decision scores [56], [57], class-inclusion probabilities with extreme value theory (EVT) [58], distance [59], [60], and so on. Discriminative methods are pioneering exploration about OSR, while they are hard to effectively constrain the potential open space and adapt to the numerous data. By contrast, generative methods can learn the decision boundary between known and unknown categories by generating samples [61], [62], having better generalization performance for rejecting the unknown categories. Furthermore, some generative methods based on reconstruction achieve better OSR performance by combining with conventional judge criteria [63], [64].

However, works mentioned above are aiming at OSR problem on daily nature images with characteristics of sufficient labels, more categories, and higher openness. As for the open-set HSI classification, the prominent characterizes are small labels, fewer categories, and lower openness. Therefore, a specific multitask deep learning method is proposed to tackle the few-shot issue in the HSI open world (MDL4OW) [65]. To further effectively utilize the spectral–spatial features of all samples, a spectral–spatial latent reconstruction (SSLR) learning method is proposed to detect the unknown categories and simultaneously improve the classification accuracy of known categories [66]. The above two methods based on reconstruction are among the few jobs for open-set HSI classification, which have pathbreaking value.

However, the methods based on reconstruction are proven to be unstable to the open-set problem, because the generative models cannot fully cognize the information of unknown categories [67]. Besides, the laborious determination of threshold weakens the adaptiveness of the methods. Obviously, two crucial points to open-set HSI classification are how to better utilize the known categories to model the extra-category space and determine a learnable threshold with high adaptability. Therefore, to improve the robustness of model and reduce the risk of open space when facing arbitrary unknown categories of ground objects, we propose a spectral–spatial multiple layer perceptron (MLP)-like network with reciprocal points learning (SSMLP-RPL) for open-set HSI classification.

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To better utilize the known categories to model the extra-category space, we introduce the RPL [68], [69] to solve the problem of open-set HSI classification. The reciprocal point is a novel concept that denotes an inverse prototype for each known category. Different from the conventional learning paradigm consisting of the softmax function with a higher risk of open space, RPL makes all known categories pushed to the periphery of the space by the corresponding reciprocal points in the training stage. Therefore, the label of a known category can be determined by the distance with its reciprocal point. While the unknown categories and reciprocal points compose the extra-category space together, the set of reciprocal points can be utilized to constrain the extra-category space. Thus, the unknown categories become distinguishable by enlarging the distance between known categories and unknown categories in high-dimensional feature space. Based on this, we further propose to utilize a learnable dynamic threshold of each known category to reject the unknown categories, avoiding the laborious determination of specific threshold.

Furthermore, the mainstream backbones represented by CNN and Transformer have been dominant in HSI classification recently, which is inseparable from their powerful capacity of feature representation learning. However, recent studies prove that the feature representation learning task can be replaced by simple MLP [70]. Besides, MLP-like networks have shown their promising power in recent studies [71], [72], [73], so also been gradually introduced to solve the problem of HSI classification. A multiscale U-shape MLP (MUMLP) that embeds the MLP layers into the U-Net is proposed to significantly decrease the parameter scale and calculation consumption, but it neglects the spatial semantics [74]. An MLP-based network composed by simple linear layers based on residual learning (HSI-ResMLP) is proposed to improve the performance of HSI classification, but it directly takes MLP layers as a simply replaceable encoder of high-dimensional feature while neglect the special HSI data attribute of long-distance spectral dependency [75]. They are one of the few pioneering works that demonstrate the potential of MLP-like architecture for HSI classification. However, as initial exploratory works, they still have shortcomings in using fully connected (FC) layers to better extract joint spatial–spectral features. Therefore, aiming at better modeling discriminative dependency of spectrum and capturing conducive spatial semantics, motivated by the huge potential of MLP backbone, we construct an SSMLP-like feature extraction framework mainly composed by a series of simple FC layers to enhance the representation capacity of latent feature. SSMLP is mainly constructed by multiple SSMLP layers, and each SSMLP layer is composed by SpeFC and SpaFC two modules. Among them, through taking each spectrum as a processing chunk of FC layer, the SpeFC module enables to capture whole spectrum information and model long-distance spectral dependence. Also, the SpaFC module enables to model spatial semantics jointly in horizontal and vertical pathways. The extraction of robust spectral–spatial feature further guarantees the robust identification of known and unknown categories.

Finally, the main contributions of our work can be summarized as follows.

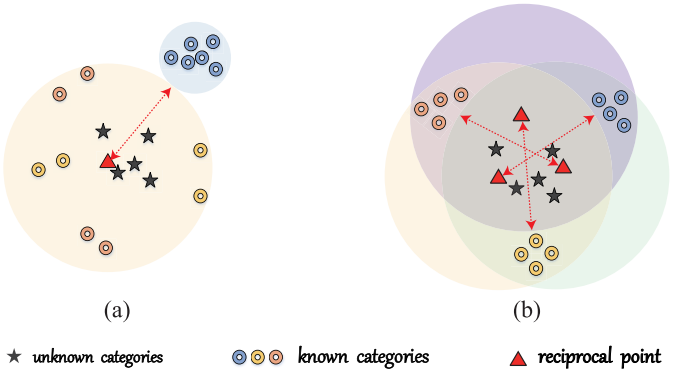


Fig. 2. Illustration of the RPL. (a) Reciprocal point for one known category. (b) Mutual exclusion between all known categories and the reciprocal points.

1) Aiming at modeling the extra-category space by the known categories to reduce the risk of open space, we introduce RPL to solve the problem of open-set HSI classification. As far as we know, it is the first work to apply novel RPL in HSI processing. Besides, we further propose a strategy of learnable dynamic threshold to improve open performance of the model effectively, and the laborious determination of specific threshold on previous works can be avoided.
2) Aiming at enhancing the representation capacity of latent feature, we propose an SSMLP-like network to capture the spectral–spatial feature merely with a series of simple FC layers, which mainly involve SpeFC and SpaFC two modules. SSMLP is capable of extracting robust feature conducive to constructing fine feature space and identifying the known and unknown categories. Besides, as an end-to-end HSI classification framework of MLP-like backbone, SSMLP network can compete with the advanced works based on CNN and Transformer.
3) Extensive experiments on three benchmark HSIs show that the proposed SSMLP-RPL has a competitive performance both in CSE and OSE and even surpasses the currently advanced closed-set and open-set HSI classification methods. Therefore, SSMLP-RPL is more robust and effective for reducing open risk in the realistic open HSI world.

Sections II–IV are as follows. Section II describes SSMLP-RPL. Section III is experimental verification and analysis. Section IV concludes this article comprehensively.

II. Proposed Method

A. Reciprocal Points Learning

The reciprocal point [68] is a novel concept that denotes an inverse prototype for each known category, so reciprocal points can be utilized to constrain the extra-category space. Also, RPL [69] makes all known categories pushed to the periphery of the space by the corresponding reciprocal points in the training stage. While the unknown categories and reciprocal points compose the extra-category space together, it makes the unknown categories become distinguishable. Therefore, RPL

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is proven to be an effective tool to minimize the overlap of known distribution and unknown distributions without loss of known classification accuracy.

Different from the conventional learning paradigm consisting of the softmax function with a higher risk of open space, we directly introduce the paradigm of RPL to better utilize the known categories to model the extra-category space and solve the problem of open-set HSI classification. Based on the original RPL, we further propose a strategy of learnable dynamic threshold to improve open-set HSI classification. In this section, the RPL for hyperspectral data instantiation will be elaborated first. Also, the improved strategy of learnable dynamic threshold will be described in Sections II-A and II-B.

Specifically, to normalize this problem, given an HSI data $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote height, width, and number of bands, respectively. Through cutting the original data $\boldsymbol{I}$, HW data cubes $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C}$ are obtained, where S denotes the patch size. Therefore, given training dataset $\mathcal{D}_{\mathcal{L}} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ with n labeled samples and N known categories, so $y_i \in \{1, \ldots, N\}$ is the corresponding label of $\boldsymbol{x}_i$. As for the test data $\mathcal{D}_{\mathcal{T}} = \{\boldsymbol{t}_1, \ldots, \boldsymbol{t}_m\}$, the labels of $\boldsymbol{t}_i$ are in $\{1, \ldots, N\}$ for closed-set HSI classification and in $\{1, \ldots, N\} \cup \{N+1, \ldots, N+U\}$ for open-set HSI classification, where U denotes the number of unknown categories in the realistic HSI scene. Thus, the potentially unknown data can be represented as $\mathcal{D}_{\mathcal{U}}$. For each known category $k (1 \le k \le N)$, the corresponding open space $\mathcal{O}_k$ in D-dimensional embedding space $\mathbb{R}^D$ can be represented as $\mathcal{O}_k = \mathbb{R} - \mathcal{S}_k$, where $\mathcal{S}_k$ denotes the embedding space of k.

Therefore, as an inverse prototype of known category k, the reciprocal point $\mathcal{P}^k$ can be regarded as a latent representation of subdataset $\mathcal{D}_{\mathcal{L}}^{\neq k} \cup \mathcal{D}_{\mathcal{U}}$. As shown in Fig. 2(a), the reciprocal point $\mathcal{P}^k$ should be closer to open space $\mathcal{O}_k$ and simultaneously far away with $\mathcal{S}_k$, which can be formulated as follows:

$$\max\left(\zeta\left(\mathcal{D}_{\mathcal{L}}^{\neq k} \cup \mathcal{D}_{\mathcal{U}}, \mathcal{P}^k\right)\right) \le d \quad \forall d \in \zeta\left(\mathcal{D}_{\mathcal{L}}^{k}, \mathcal{P}^k\right) \tag{1}$$

where ζ denotes a set of distances. Based on this, the category about each sample of $\mathcal{D}_{\mathcal{L}}$ can be determined by the distance with the set of reciprocal points $\mathcal{P}$, and the known category k has the maximal distance with $\mathcal{P}^k$.

Therefore, one of the most crucial purposes for RPL is to enlarge the distance between the known categories and their corresponding reciprocal points. Specifically, as common metrics used to measure the similarity of vectors or high-dimensional features, the Euclidean distance d_e and dot product d_d are utilized to jointly represent the distance d from the perspective of space location and angle direction, which can be formulated as follows:

$$d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) = \frac{1}{D} \bullet \left\|\varphi(\boldsymbol{x}_i) - \mathcal{P}^k\right\|_2^2 \tag{2}$$

$$d_d(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) = \varphi(\boldsymbol{x}_i) \bullet \mathcal{P}^k \tag{3}$$

$$d(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) = d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) - d_d(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) \tag{4}$$

where $\varphi(\cdot)$ denotes the feature representation network with learnable parameters set Θ.

After describing the distance metric, because a greater distance $d(\varphi(\boldsymbol{x}_i), \mathcal{P}^k)$ will lead to a larger probability for labeling the $\boldsymbol{x}_i$ with the category k, the softmax function can be utilized to calculate the category probability following the property of reciprocal point, which can be formulated as follows:

$$p(y = k|\boldsymbol{x}_i, \varphi, \mathcal{P}) = \frac{e^{\gamma d(\varphi(\boldsymbol{x}_i), \mathcal{P}^k)}}{\sum_{j=1}^{N} e^{\gamma d(\varphi(\boldsymbol{x}_i), \mathcal{P}^j)}}. \tag{5}$$

Therefore, the supervised loss function can be defined as the negative log probability of the labels k

$$\mathcal{L}_C(\boldsymbol{x}; \Theta, \mathcal{P}) = -\log p(y = k|\boldsymbol{x}_i, \varphi, \mathcal{P}). \tag{6}$$

The learnable parameters set Θ of feature representation network $\varphi(\cdot)$ is updated by minimizing the $\mathcal{L}_C$. Thus, $\varphi(\cdot)$ obtains the capability of classifying the known categories through supervised learning with multiclass samples and their corresponding true labels. More importantly, as shown in Fig. 2(b), all known categories are pushed to the periphery of the space by the corresponding reciprocal points in the training stage, and the reciprocal points belonging to the extra-category space are also pushed to be separated from the known space.

However, the potential unknown categories also belong to the extra-category space, so the risk of open space still exists. To reduce the risk of open space and make the unknown category more distinguishable, the open space should be further constrained. Therefore, the reciprocal points are utilized to constrain the unknown space based on (1), which can be formulated as follows:

$$\max\left(\zeta\left(\mathcal{D}_{\mathcal{L}}^{\neq k} \cup \mathcal{D}_{\mathcal{U}}, \mathcal{P}^k\right)\right) \le R \tag{7}$$

where R is the constrained boundary to restrict the potential unknown categories. Because the spaces $\mathcal{S}_k$ and $\mathcal{O}_k$ are complementary to each other, the boundary R can be learned indirectly by enlarging the distance $d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k)$ to be larger than R with the margin loss function

$$\mathcal{L}_O(\boldsymbol{x}; \Theta, \mathcal{P}^k, R) = \max(d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k) - R, 0) \tag{8}$$

where $\mathcal{L}_O$ is an unsupervised loss function.

The learnable parameters set Θ of feature representation network $\varphi(\cdot)$ is also updated by minimizing the $\mathcal{L}_O$. As the $d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k)$ is enlarged based on (6), the boundary R also increases and keeps approaching $d_e(\varphi(\boldsymbol{x}_i), \mathcal{P}^k)$ based on (8). Therefore, the distance between the potential unknown categories and the reciprocal points is constrained to the range of boundary R through minimizing the $\mathcal{L}_O$. The constraint boundary R is determined by the optimization of network when training. After finishing training, the fit network projects the features to the high dimension to construct the feature space, and the boundary R is also determined indirectly at the same time.

Through the interaction based on (6) and (8), the distance between known categories and unknown categories is enlarged in high-dimensional feature space through RPL, which makes it possible to identify the unknown categories.

B. SSMLP-Like Network

Although the novel learning paradigm of reciprocal points can produce exquisite embedding space for open-set HSI

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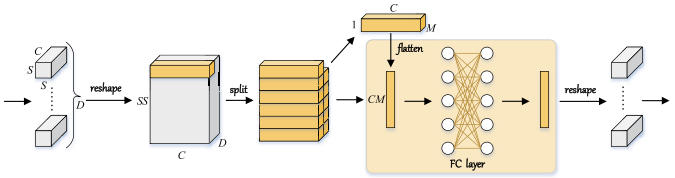


Fig. 3. Illustration of the SpeFC module. SpeFC module can model spectral feature by directly taking spectral chunks as processing unit for the FC layer; thus, the long-distance spectral information can be effectively modeled merely with a simple FC layer.

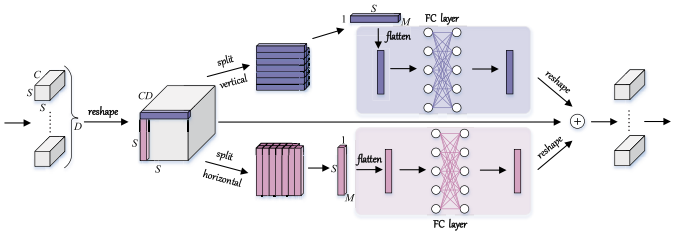


Fig. 4. Illustration of the SpaFC module. SpaFC module can model the spatial semantic feature jointly along the vertical and horizontal directions by leveraging multibranch FC layers.

classification, the extraction of discriminative feature is still crucial for robust identification [76], while the extraction of powerful feature depends on the performance of feature representation network to a large extent. To enhance the capacity of feature learning, an SSMLP-like network is proposed to capture the latent feature representation merely with a series of FC layers. The cores of SSMLP involve SpeFC and SpaFC two modules, which are designed to model the long-distance spectral information and large-range spatial semantics.

It should be noted that the channel-MLP and permute-MLP in the vision permutator (ViP) [73] are designed for three-channel natural images; thus, they both take spatial dimension as one special dimension and directly project the channel dimension into the high dimension, which are similar to the 2-D convolution layer to some extent. However, the SpeFC and SpaFC modules in SSMLP-like network are designed for high-dimensional HSI data; thus, they both take the spectral dimension as one special dimension and project whole 3-D HSI cube into the high dimension, which are similar to the 3-D convolution layer to some extent.

As for HSI data $\boldsymbol{I} \in \mathbb{R}^{H\times W\times C}$ and HW data cubes $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C}$, each cube $\boldsymbol{x}$ is taken as the input of the SSMLP and belongs to one category of the ground objects.

1) SpeFC Module: Aiming at the long-distance spectral information contained in HSI data, we design the SpeFC module to model spectral feature by taking spectral chunks as processing unit for the FC layer, which is also a noteworthy distinction with the recent works of MLP-like network in image processing [71], [72], [73].

Specifically, as shown in Fig. 3, after projecting the D-dimension, $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C\times D}$ is as the input for the SpeFC Module. FC layer usually has a huge computation cost with a high-dimensional feature. Following the general operation [73], we first split $\boldsymbol{x}$ along the D-dimension with a segmentation ratio M to obtain D/M groups of tokens $\boldsymbol{x}^q \in \mathbb{R}^{S\times S\times C\times M}$ $(q = 1, \ldots, D/M)$ to achieve indirect dimension reduction. For each pixel in cube $\boldsymbol{x}$, the full spectral feature with M-dimension is taken as a chunk $\boldsymbol{x}_j^q \in \mathbb{R}^{C\times M}$ $(j = 1, \ldots, S^2)$ to model the long-distance spectral information. Then, every chunks will pass an FC layer with weight $\boldsymbol{\mathcal{W}} \in \mathbb{R}^{CM\times CM}$, which can be formulated as follows:

$$\hat{\boldsymbol{x}}_j^q = \boldsymbol{x}_j^q \boldsymbol{\mathcal{W}}. \tag{9}$$

At last, all chunks $\hat{\boldsymbol{x}}_j^q \in \mathbb{R}^{C\times M}$ are reshaped back to dimension of the original tokens to obtain $\hat{\boldsymbol{x}} \in \mathbb{R}^{S\times S\times C\times D}$. Therefore, the long-distance spectral information can be effectively modeled merely with a simple FC layer. Compared with the local sliding window of convolution operation, the designed SpeFC Module with FC layer is obviously better at establishing long-distance dependence on spectral dimension. It better contributes to enhance the representation of spectral feature.

2) SpaFC Module: Because of a dilemma between the spatial correlation in HSI data and the in-line structure of FC layer, it is difficult to capture valuable spatial semantic merely with a simple FC layer. Therefore, we refer to the ViP [73] designed for image processing to leverage multibranch FC layers to model spatial information jointly.

Specifically, as shown in Fig. 4, $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C\times D}$ is as the input for the SpaFC Module. We first split $\boldsymbol{x}$ along the vertical direction and obtain S chunks $\boldsymbol{x}_l \in \mathbb{R}^{S\times C\times D}$ $(l = 1, \ldots, S)$. Similarly, we split $\boldsymbol{x}_l$ along the D-dimension with a segmentation ratio M to obtain D/M groups of tokens $\boldsymbol{x}_l^q \in \mathbb{R}^{S\times C\times M}$ $(q = 1, \ldots, D/M)$ to reduce computation cost. For each band in $\boldsymbol{x}_l^q$, the M-dimensional spatial feature along the vertical direction is taken as a chunk $\boldsymbol{x}_l^{qn} \in \mathbb{R}^{S\times M}$ $(n = 1, \ldots, C)$ to pass an FC layer with weight $\boldsymbol{\mathcal{W}} \in \mathbb{R}^{SM\times SM}$, which can be formulated as follows:

$$\hat{\boldsymbol{x}}_l^{qn} = \boldsymbol{x}_j^{qn} \boldsymbol{\mathcal{W}}. \tag{10}$$

Then, all chunks $\hat{\boldsymbol{x}}_l^{qn} \in \mathbb{R}^{S\times M}$ are reshaped back to dimension of the original tokens to obtain $\hat{\boldsymbol{x}} \in \mathbb{R}^{S\times S\times C\times D}$. As for the horizontal branch, we split $\boldsymbol{x}$ along the horizontal direction and obtain S chunks $\boldsymbol{x}_l \in \mathbb{R}^{S\times C\times D}$ $(l = 1, \ldots, S)$. Other operations do likewise above, which achieves modeling the spatial feature along the horizontal direction. Besides, an FC layer with weight $\boldsymbol{\mathcal{W}} \in \mathbb{R}^{D\times D}$ is utilized to process the D-dimensional tokens directly to achieve an interaction between features of different dimensions. At last, outputs of three branches execute an elementwise summing to achieve modeling spatial semantic jointly.

3) Overall Structure: The overall structure of SSMLP network is shown in Fig. 5. Actually, almost all current works of MLP-like architecture require to project the feature into high dimension by a preset embedding layer. The preset embedding layer involves the convolution layer [72] or simple linear layer [73]. Among them, the projection through a linear layer is a simple and direct method, while the embedding layer based on the convolution can better combine the superiority of convolution capturing spatial semantic.

Therefore, in the SSMLP network, each HSI data cube $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C}$ first requires to be projected into the high dimension, and an embedding layer is set to achieve the projection by

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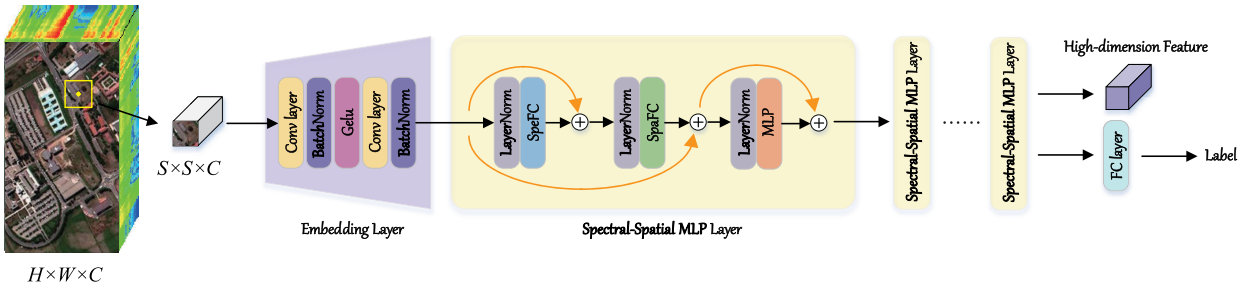


Fig. 5. Illustration of the overall structure for SSMLP network. SSMLP network is mainly composed by an embedding layer and multiple SSMLP layers. Among them, the embedding layer is responsible for projecting the feature into high-dimensional space, and the high-dimensional feature after projection is input into multiple SSMLP layers to model latent representation. At last, the latent representation of feature is output for constructing embedding space by RPL. Simultaneously, an additional FC layer can produce a distance between samples and the reciprocal points to obtain a corresponding label.

deploying convolution layers. Two layers of 3-D convolution are set to model HSI data cubes, and the convolution layer is followed by a layer of batch normalization and the Gaussian error linear unit (GeLu) activation function. The convolutional kernel size of first layer is (3, 1, 1), and its dimension of projection is from C to $D/2$, which is set to mainly model spectral information, where D denotes the dimension of final projected feature. The convolutional kernel size of second layer is (1, 3, 3), and its dimension of projection is from $D/2$ to D, which is set to mainly model spatial information.

Then, the high-dimensional feature $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C\times D}$ after projection is input into multiple SSMLP layers to model latent representation. As for each SSMLP layer, a SpeFC module, a SpaFC module, and a basic MLP layer are contained as sequence, and the layer normalization is deployed before each module. Besides, residual skip connections are also applied to make stable for the deep network. Therefore, except for the embedding layer, SSMLP network contains L SSMLP layers, and L directly determines the depth of the network. At last, the high-dimensional feature is output for constructing embedding space by RPL. Simultaneously, an additional FC layer can produce a distance between samples and the reciprocal points to obtain a corresponding label.

Algorithm 1 Learning for Open-Set HSI Classification

Input: The HSI data cubes $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C}$ with true label y for training; The spectral–spatial MLP (SSMLP) network $\varphi(\cdot)$ with initialized parameters Θ; The reciprocal points $\mathcal{P}$ and the constrained boundary R;
Output: The optimized parameters Θ, $\mathcal{P}$, and R.
1: while $\varphi(\cdot)$ *is not converge* do
2: Calculate the gross loss by $\mathcal{L} = \mathcal{L}_C + \lambda\mathcal{L}_O$ (Eq. 11);
3: Update the parameters Θ, $\mathcal{P}$, and R by the backpropagation algorithm;
4: end while
5: **return** $\varphi(\cdot)$, $\mathcal{L}_O$;
Input: An HSI data cube $\boldsymbol{x} \in \mathbb{R}^{S\times S\times C}$ for testing;
Output: The predicted open-set label $\hat{y}$.
6: Calculate the distance vector $\varphi(\boldsymbol{x}) = \boldsymbol{d} \in \mathbb{R}^{1\times N}$ between $\boldsymbol{x}$ and $\mathcal{P}$;
7: Predict the closed-set label $\hat{y} = \text{argmax}(\varphi(\boldsymbol{x}))$;
8: Calculate the dynamic threshold $\mathcal{T}$ by Eq. 13;
9: if $\boldsymbol{d}(\hat{y}) < \mathcal{T}$ then
10: $\hat{y}$ = unknown;
11: end if
12: **return** $\hat{y}$.

C. Learning for Open-Set HSI Classification

In the above two subsections, two significant parts of SSMLP-RPL are described, respectively, involving RPL and SSMLP network. But, there are still two key questions, that are, optimization of the network and specific identification of the unknown categories, which will be further described in this section. As shown in Algorithm 1, first, according to (6) and (8), the gross loss function should combine two functions to minimize the loss jointly

$$\mathcal{L}(\boldsymbol{x}; \Theta, \mathcal{P}, R) = \mathcal{L}_C(\boldsymbol{x}; \Theta, \mathcal{P}) + \lambda\mathcal{L}_O(\boldsymbol{x}; \Theta, \mathcal{P}, R) \quad (11)$$

where λ is a trade-off weight that controls the contribution of second term. Minimizing $\mathcal{L}$ implements the iterative optimization of SSMLP network and feature embedding space. After training, the SSMLP network is capable of pushing the known categories to the periphery of the embedding space by the corresponding reciprocal points and restricting the unknown categories in a constrained range simultaneously, which makes the unknown categories more distinguishable. Next, a specific threshold should be determined to identify the unknown categories.

A naive attempt is to simply take the learned boundary R in embedding space as a threshold according to the basic assumption of RPL; that is, all unknown categories are constrained in a range with the boundary R. However, we find the distance in (8) between a known category and its reciprocal point is not always consistent for all known categories, which can be formulated as follows:

$$d_e(\varphi(\boldsymbol{x}_k), \mathcal{P}^k) \neq d_e(\varphi(\boldsymbol{x}_{\neq k}), \mathcal{P}^{\neq k}), \quad (R \leq \min(d_e)). \quad (12)$$

Simply taking the boundary R as the threshold may lead to lose some potential unknown categories that are closer to the known category with a larger d_e. Therefore, the fixed boundary R cannot well adapt to the complex and changing distribution of multiple unknown categories.

Aiming at this problem, we propose to utilize a dynamic threshold $\mathcal{T}$ for different known categories. Specifically, according to (8), $\mathcal{L}_O$ can be considered as the difference

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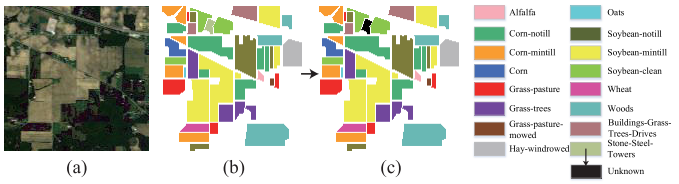


Fig. 6. IP. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

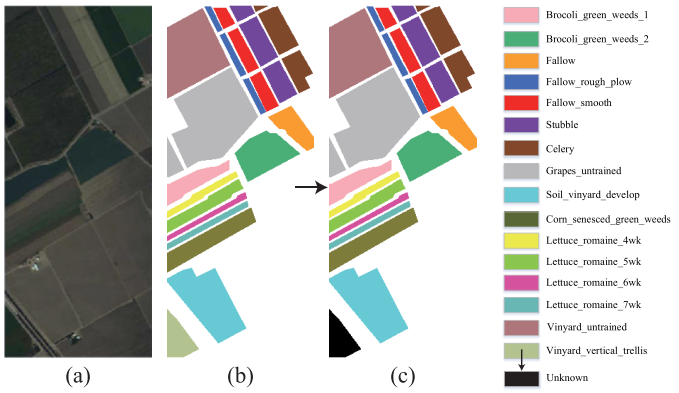


Fig. 7. SA. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

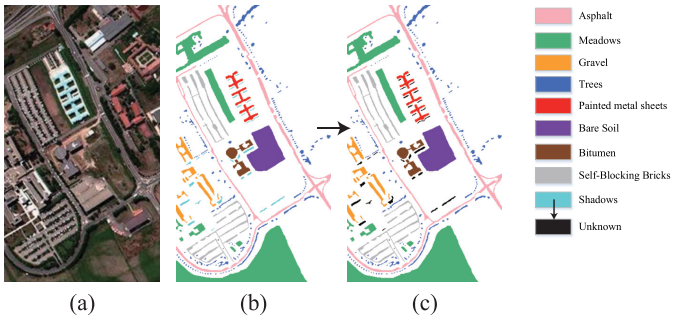


Fig. 8. PU. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

between the minimal values of all d_e and R, so we utilize the final $\mathcal{L}_O$ to represent the dynamic threshold $\mathcal{T}$

$$\mathcal{T}^k = \min(d_e(\varphi(\boldsymbol{x}_k), \mathcal{P}^k)) - \mathcal{L}_O \tag{13}$$

where $\boldsymbol{x}_k$ here denotes the training sample with a true label k. Compared with the fixed threshold R, the dynamic threshold can better identify some unknown categories that are closer to the known category with a larger d_e. Also, it is learnable during training, which also avoids the laborious handcraft setting of previous open-set HSI classification methods. After determining the set of dynamic threshold, a sample with the distance d_e smaller than the threshold will be identified as the unknown categories. Therefore, the proposed strategy of learnable dynamic threshold can maximize the rejection of unknown categories without affecting the identification of known categories as far as possible.

TABLE I
DETAILED PROPERTIES OF EXPERIMENTAL HSI DATA

Data Name	Indian Pines (IP)	Salinas (SA)	Pavia University (PU)
Image Size	145×145	512×217	610×340
Number of Bands	200	204	103
Sensor	AVIRIS	AVIRIS	ROSIS
Spectral range	0.40-2.50 μm	0.40-2.50 μm	0.43-0.86 μm
Spatial resolution	20 m	3.7 m	1.3 m

III. EXPERIMENTAL RESULTS AND ANALYSIS

The environmental hardware is a workstation equipped with the Intel Core Intel[1] Xeon[1] Gold 5218 processer, 256-GB RAM, and Nvidia A100 PCIE GPU. Also, Python and PyTorch library are utilized to finish the proposed program.

A. Datasets

To conduct the experiment of open-set HSI classification and demonstrate the proposed method comprehensively, three familiar HSI data are selected: Indian Pines (IP), Salinas (SA), and Pavia University (PU). Their details and parameters are shown in Table I. Their false-color images are shown in Figs. 6(a), 7(a), and 8(a). Their standard ground truth for closed-set HSI classification are shown in Fig. 6(b), 7(b), and 8(b). To utilize the acknowledged and standard ground truth to simulate the condition of open-set HSI classification, we uniformly take the last category of ground objects as the unknown category for fairness. Simultaneously, setting only one unknown category also ensures the overall effect of the open-set classification maps for a detailed analysis. So, their open-set ground-truth maps are shown in Figs. 6(c), 7(c), 8(c). The settings above are the default. If the specific setting of the unknown categories changes in subsequent experiments, we will restate it in the corresponding subsection.

Besides, to simulate the practical condition of a limited number of training samples, we randomly select 30 labeled samples from each category of ground objects to compose the training set (for IP data, only ten labeled samples are selected for the seventh and ninth categories because of the few samples). The rest of all labeled samples are taken as the test set.

B. Implementation Details

A series of default settings will be described in this section. The trade-off weight λ for $\mathcal{L}_O$ in (11) is set to 0.1. All parameters of the SSMLP network are randomly initialized before training. The initial learning rate is 0.001, and the number of training iterations is set to 100. The projected dimension D is set to 64, and the segmentation ratio M is set to 8. The optimal number L of SSMLP layers and the spatial patch size S of a cube will be determined in the subsequent experiments. Besides, a common data augmentation strategy is implemented to mitigate the dilemma of small samples, which refers to spatially flipping the cubes from different directions to expand the size of training data. All trials will

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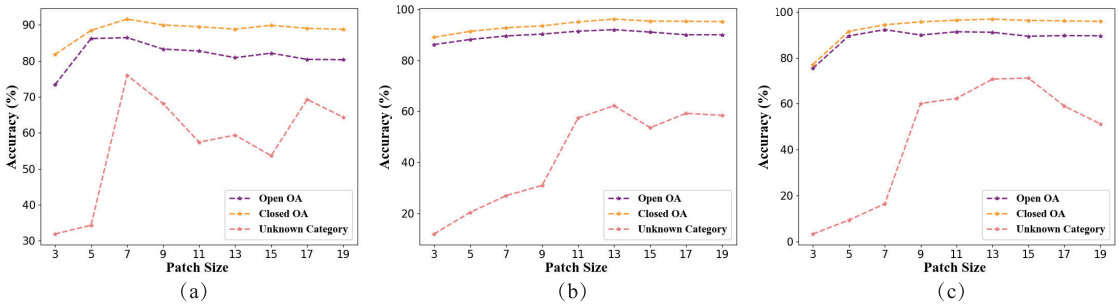


Fig. 9. Line charts of performance varying with the change of patch size S. (a) IP data. (b) SA data. (c) PU data.

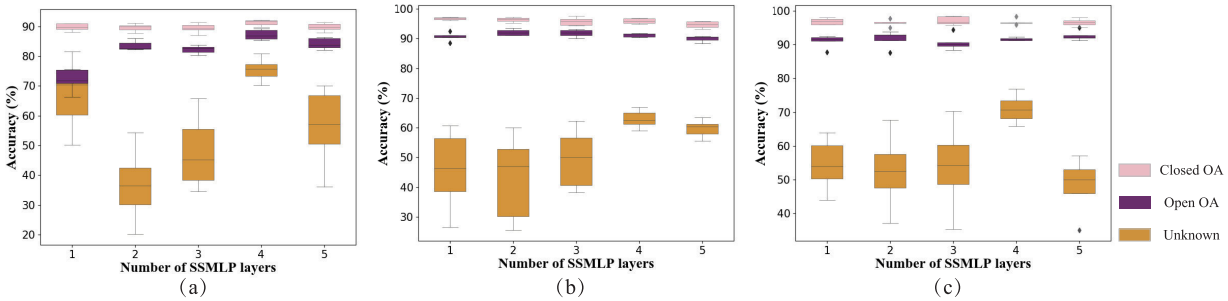


Fig. 10. Boxplot graphs of performance varying with the number of SSMLP layers L. (a) IP data. (b) SA data. (c) PU data.

be conducted ten times with different combinations of random training samples to eliminate experimental errors farthest. The average results are taken for analysis and comparison.

As for the evaluation metrics, we still choose several common metrics for HSI classification, which involve the accuracy per category, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Since the experiments focus on open-set HSI classification, these metrics are transformed to be open by calculating them both on the known and unknown categories. Furthermore, the closed-set metrics are also calculated to verify the routine performance of methods with the assumption of a closed environment.

C. Analysis of Patch Size

The spatial patch size S of HSI data cubes will lead to a significant impact on the final performance, because the S directly determines the spatial range that the MLP layers can perceive. Thus, we implement experiments on three HSI data to observe the performance varying with the change of patch size S and determine the optimal S. The results are shown in Fig. 9. As we can observe, the performance usually increases with the patch size S expanding on SA and PU data having a larger image size but fluctuates on IP data having a smaller image size. For the balance of performance and calculation consumption, we set S to 7 for IP data and set S to 13 for SA and PU data.

D. Analysis of Network Depth

Depth of the network directly determines the performance of the model. Although deepening the network depth can probably obtain a better performance, a larger number of trainable parameters are easy to lead to a degradation of model performance under the condition of a limited size of samples. Therefore, we implement experiments on three HSI data to observe the performance varying with the change of the number L of SSMLP layers and determine the optimal L. The results are shown in Fig. 10. As we can observe, the closed performance is almost unaffected by the deepening of the network, but the accuracy of the unknown category and open performance is higher and more stable with the number of SSMLP layers increasing. Therefore, we finally set the optimal L to 4 to determine the depth of the SSMLP network.

E. Experiment for Comparison

To comprehensively verify the performance of the proposed method for open-set HSI classification, we select several influential methods for comparison.

First, comparing with the conventional closed-set HSI classification methods is necessary to verify the superiority of open-set classification methods when facing the unknown categories. Simultaneously, although the conventional methods are designed with an assumption of CSE, they also can achieve the open-set classification through some specific improvements [76]. Therefore, we select several mainstream state of the art (SOTA) HSI classification methods that are pretty representative, including one method based on the classifier of support vector machine (SVM): deep few-shot learning (DFSL) [49]; two methods based on CNN backbone: double-branch dual-attention network (DBDA) [24] and 3-D octave convolution with spatial–spectral attention network (3DOC-SSAN) [26]; two methods based on Transformer backbone: spectral–spatial Transformer network (SSTN) [34] and spectral–spatial feature tokenization Transformer (SSFTT) [37]. As for the DFSL, it cannot predict

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TABLE II

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE IP SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 16 CATEGORY IS DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
1	100.0±0.0	92.17±5.76	97.39±1.62	98.47±2.18	99.56±0.86	100.0±0.0	100.0±0.0	100.0±0.0
2	64.69±5.72	34.80±8.29	68.60±5.19	37.43±9.31	81.48±3.18	88.12±2.87	87.61±1.54	88.18±3.21
3	85.73±4.35	20.84±13.28	70.30±4.44	53.96±8.44	85.69±8.41	89.18±3.85	91.75±1.99	87.75±4.61
4	98.39±1.88	51.09±14.41	91.64±4.22	82.40±7.75	98.27±1.37	94.81±2.70	91.56±0.42	95.65±3.04
5	91.90±2.69	56.35±8.83	85.50±2.64	62.79±7.67	90.12±2.73	89.17±2.75	88.20±7.45	92.60±1.77
6	96.02±2.74	70.17±13.32	92.54±4.29	83.87±5.09	97.45±1.54	97.68±2.61	99.59±0.41	95.51±3.67
7	100.0±0.0	48.21±31.75	93.57±7.28	89.28±7.31	100.0±0.0	89.64±1.07	89.29±10.71	97.62±2.11
8	97.88±1.34	88.87±5.41	97.48±1.50	94.56±2.30	99.89±0.25	99.87±0.21	98.12±0.63	98.53±1.56
9	100.0±0.0	33.5±39.68	91.01±5.38	82.50±10.06	98.50±2.29	100.0±0.0	97.50±2.50	95.31±1.23
10	77.11±4.04	47.90±10.47	74.32±3.22	53.66±7.96	88.33±4.09	80.97±2.90	70.88±7.39	87.05±2.14
11	66.38±3.88	38.32±11.57	58.63±3.90	45.88±9.85	74.22±5.36	69.13±4.79	31.01±7.01	83.64±3.90
12	68.19±4.32	38.98±9.40	86.50±5.35	52.61±7.05	95.41±2.56	88.68±3.13	69.81±2.36	93.23±3.21
13	98.92±1.08	98.34±3.05	98.24±1.13	95.46±5.35	99.65±0.53	99.76±0.45	99.76±0.24	97.14±1.26
14	92.65±2.47	76.86±10.68	93.04±2.15	78.19±6.23	96.23±1.56	95.49±1.35	91.30±0.40	93.29±1.21
15	93.49±3.77	44.76±17.41	88.80±5.12	82.51±6.32	97.15±2.01	91.30±4.13	88.60±7.25	94.43±3.21
Unknown	0.0±0.0	4.62±10.31	17.20±23.59	13.76±8.59	9.56±17.60	96.88±2.65	**99.46**±0.54	75.74±5.76
Closed OA	79.70±1.29	65.19±3.33	89.53±1.39	62.21±2.16	91.44±1.22	90.71±1.26	85.68±1.61	**91.60**±1.24
Closed AA	88.76±0.86	71.59±4.60	91.98±0.61	75.11±1.87	93.85±0.54	93.26±0.66	87.32±1.24	**94.37**±1.46
Closed Kappa	77.09±1.44	60.88±3.72	87.40±1.55	57.74±2.22	89.30±1.37	88.54±1.88	83.29±2.07	**89.59**±1.07
Open OA	78.98±1.28	49.28±5.62	76.46±1.14	59.59±2.15	83.62±1.79	85.24±1.34	84.15±1.24	**86.45**±1.81
Open AA	83.21±0.81	52.86±7.25	81.55±1.46	69.21±1.87	85.22±1.37	89.79±0.73	85.15±1.13	**90.42**±2.51
Open Kappa	76.31±1.43	45.51±5.57	74.13±1.19	55.26±2.18	83.02±1.96	84.85±1.48	81.30±1.32	**85.70**±1.99

a novel category because of the closed property of SVM. As for other methods, they commonly deploy an FC layer as the score layer at the tail of the network to predict scores and take the index with the maximal score as the final label. Following a general solution for the open-set problem [76], we uniformly deploy an additional softmax function for their networks, so the index with the maximal probability (actually consistent with the index with maximal score) will be taken as the predicted label in the probability vector. We refer to the general solution [76] and set the threshold of softmax function to 0.5 to identify the unknown categories (i.e., the output with the maximal probability smaller than 0.5 will be taken as the unknown category).

Second, a comparison with the specialized open-set HSI classification methods is also necessary. At present, there are few open-set related works, which mainly involve two reconstruction-based methods: multitask deep learning method for open world (MDL4OW) [65] and SSLR learning [66]. Therefore, we select both for comparison. A common characteristic of the both is they take the pixels whose values of reconstruction loss function are larger than the threshold as the unknown categories.

The settings of all comparative methods are referred to the optimal setting of each original literature, and the combinations of random training samples are consistent for all methods in ten times of trials to guarantee fairness. It should be noted that the closed-set and open-set performance of each method will be verified simultaneously by recording the open and closed metrics. According to the assumption of the unknown category in Section III-A, the experimental results are shown in Tables II–IV. Based on the observation of results, several valuable conclusions are obtained as follows.

1) The closed-set methods with a score layer generally classify all instances to different known categories. We make them capable of identifying the unknown category by deploying a softmax function on the tail of the network. But, this practice cannot be applied to the SVM-based methods represented by DFSL because of the completely closed property of SVM. Thus, the DFSL's identification accuracy of the unknown category is 0, and DFSL still classifies every instance of the unknown category into different known categories. As for other closed-set methods, this practice effectively improves their identification performance on the unknown categories. However, even such four improved closed-set methods are capable of identifying the potential unknown category, their performance is not stable with a huge standard deviation and a lower accuracy of unknown category. It denotes the inherent limitation of closed-set classifiers in OSE.
2) No matter the conventional classification methods or the specialized open-set classification methods, their performance metrics in OSE are all lower than that in CSE, which denotes that all methods accompany a performance degradation when facing the unknown category. However, the degree of performance degradation indicates the model's open-set classification performance. In general, the smaller the degree of degradation, the model will be more stable when facing the unknown categories. Also, three specialized open-set methods with a smaller degree of performance degradation clearly perform better than conventional classification methods in OSE.
3) No matter the conventional classification methods or the specialized open-set classification methods, their open metrics are strongly associated with the closed metrics. It denotes that a method with advanced open-set performance inevitably accompanies an excellent closed-set performance, while the reverse is not valid. For instance, three specialized open-set methods with better open-set

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TABLE III

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE SA SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 16 CATEGORY IS DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
1	98.41±1.35	96.74±3.39	99.16±0.57	79.64±23.45	99.84±0.15	92.68±0.72	96.53±2.49	99.45±1.02
2	99.29±0.42	83.62±22.66	99.97±0.06	72.71±21.16	99.99±0.01	96.20±0.71	98.58±0.62	98.17±1.53
3	99.54±0.24	55.06±35.76	99.61±0.78	74.26±24.15	99.99±0.01	99.64±0.47	99.04±0.98	99.60±0.25
4	98.89±1.16	79.07±38.25	98.79±0.58	94.89±3.50	98.97±1.13	98.77±1.26	98.26±1.32	98.55±0.77
5	98.68±0.76	79.24±23.21	97.10±1.31	86.26±4.38	96.77±2.05	91.66±2.16	98.48±1.40	98.70±1.03
6	98.96±2.07	99.57±0.55	98.55±0.87	95.78±2.82	99.66±0.36	97.39±2.17	97.11±2.35	94.63±3.56
7	99.39±0.54	98.35±1.59	99.48±0.22	90.26±10.21	99.70±0.22	96.26±0.59	95.42±1.09	96.33±1.23
8	76.16±3.93	49.73±30.43	68.25±6.11	43.68±25.70	81.11±4.36	78.60±13.79	76.75±9.56	87.60±2.06
9	94.46±2.62	89.05±29.70	99.18±0.62	83.78±10.46	99.89±0.10	99.57±0.43	98.96±0.51	98.34±1.12
10	91.76±2.64	50.30±30.87	88.40±3.92	70.18±7.49	95.47±0.99	96.44±1.02	95.83±1.63	94.97±2.46
11	96.94±1.17	75.03±31.51	95.57±2.87	88.66±9.12	99.08±0.80	98.20±0.45	94.07±2.10	98.48±0.65
12	99.94±0.14	48.37±48.46	99.35±0.33	89.11±6.53	99.44±0.54	99.84±0.16	99.93±0.22	99.14±0.33
13	99.89±0.10	48.25±48.28	97.93±0.88	98.70±0.91	99.75±0.28	99.45±0.40	99.53±0.40	97.79±1.65
14	98.14±1.52	73.14±32.90	96.28±2.31	92.67±4.63	98.99±0.49	99.19±0.63	99.13±0.82	96.13±1.61
15	83.88±4.12	53.27±28.97	75.39±3.91	57.43±23.53	87.69±4.59	84.20±13.00	83.25±12.44	89.28±1.99
Unknown	0.0±0.0	17.95±15.74	23.50±17.02	17.02±12.77	21.06±13.80	31.30±10.36	47.09±8.05	**62.39**±4.17
Closed OA	91.00±1.00	76.36±5.28	93.58±0.68	73.79±5.26	94.43±0.79	92.74±2.07	91.91±1.27	**96.39**±0.72
Closed AA	95.62±0.48	80.99±7.87	96.17±0.31	82.75±3.95	96.58±0.35	96.91±0.67	96.34±0.56	**97.92**±0.46
Closed Kappa	89.96±1.11	73.69±5.85	92.82±0.76	71.05±5.67	93.77±0.89	90.62±1.29	89.21±2.28	**94.89**±1.01
Open OA	87.96±0.96	67.92±8.67	86.82±1.29	70.01±5.28	89.19±0.68	88.33±1.99	88.86±1.34	**92.08**±0.98
Open AA	89.65±0.45	68.55±12.36	88.03±1.24	77.19±3.59	90.34±0.48	90.27±0.73	90.74±1.23	**92.33**±1.19
Open Kappa	86.64±1.06	64.78±9.16	85.48±1.40	67.10±5.61	88.21±0.74	87.03±2.17	87.61±1.48	**91.17**±1.10

TABLE IV

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE PU SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 9 CATEGORY IS DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
1	87.26±2.72	53.57±21.70	86.98±3.64	68.01±3.84	89.31±1.67	93.32±2.28	88.65±4.27	87.59±6.03
2	90.81±2.92	56.47±23.00	88.51±5.15	71.55±16.95	90.85±4.46	96.59±3.12	93.61±4.01	93.57±4.20
3	86.85±3.82	34.46±19.12	87.14±3.65	59.90±5.89	90.90±4.52	90.45±3.61	90.78±3.97	90.39±5.08
4	95.60±1.16	87.60±8.55	95.36±1.90	91.71±2.78	92.86±2.26	95.99±1.35	95.50±1.37	88.73±5.81
5	100.0±0.0	91.97±19.40	99.97±0.03	97.96±2.90	99.99±0.02	81.00±7.08	94.67±4.63	99.20±2.32
6	94.52±2.57	44.23±20.02	93.76±5.95	62.59±10.71	95.86±3.01	95.25±3.35	93.43±5.64	95.40±2.87
7	94.39±1.63	62.72±25.20	97.42±2.07	73.30±6.27	99.12±0.92	96.23±1.20	94.74±1.71	96.12±2.48
8	82.37±4.77	45.64±22.28	83.89±4.30	60.50±12.84	88.02±3.12	88.77±4.98	80.81±5.91	81.82±8.14
Unknown	0.0±0.0	20.80±19.14	7.62±7.69	1.84±1.37	35.73±16.79	42.99±13.96	52.09±11.81	**70.71**±6.28
Closed OA	90.51±1.60	65.55±8.16	93.94±1.39	71.51±6.79	94.35±1.61	92.57±1.43	93.69±1.83	**96.87**±0.90
Closed AA	91.47±1.08	68.35±9.76	95.63±0.49	73.99±2.15	95.86±0.70	93.50±0.79	94.28±1.23	**96.23**±1.14
Closed Kappa	87.46±2.06	55.39±11.69	92.01±1.78	63.77±7.07	92.53±2.08	92.01±1.76	92.28±1.92	**94.92**±1.81
Open OA	88.51±1.56	55.54±17.76	87.76±1.70	69.21±6.71	90.65±2.00	88.01±1.43	89.20±1.81	**91.11**±1.78
Open AA	81.31±0.96	56.38±12.63	82.29±1.37	65.26±1.93	88.07±2.30	87.62±1.40	87.81±1.56	**89.28**±1.56
Open Kappa	84.96±1.97	47.37±16.49	84.22±2.03	61.33±6.84	87.88±2.50	86.77±1.83	87.16±2.27	**88.44**±2.24

performance all have a not bad performance in CSE, which are capable of surpassing most currently advanced works. However, as a superior closed-set method, DBDA even has a better closed-set performance than MDL4OW and SSLR on the SA and PU data, while its open-set performance cannot compete with such two specialized open-set methods.

4) As for the conventional classification methods, conclusion (3) is more distinct, because their capacity to identify the unknown category heavily relies on the capacity of feature extraction. Specifically, DFSL as a specialized few-shot method has a not bad performance under the condition of limited training samples size, while it is completely unable to identify a novel category. The 3DOC-SSAN and SSFTT proved to have a superior performance with a sufficient number of training samples in practice cannot obtain a good and stable performance under the condition of limited training samples size, so they naturally perform badly in OSE. The representative CNN-based method DBDA and Transformer-based method SSTN are capable of achieving superior closed performance because of the powerful capacity of feature extraction, which also improves their open performance. Among them, SSTN obtains the highest closed and open metrics in the conventional classification methods, which even surpasses two specialized open-set methods MDL4OW and SSLR on the SA and PU data. Therefore, we can obtain another highly valuable conclusion: a superior closed-set network also has a not bad performance in open-set HSI classification by simply deploying a softmax function (a default threshold of 0.5) on the tail of the network. But, this practice cannot be effective in rejecting the unknown categories, which corresponds to conclusion (1).

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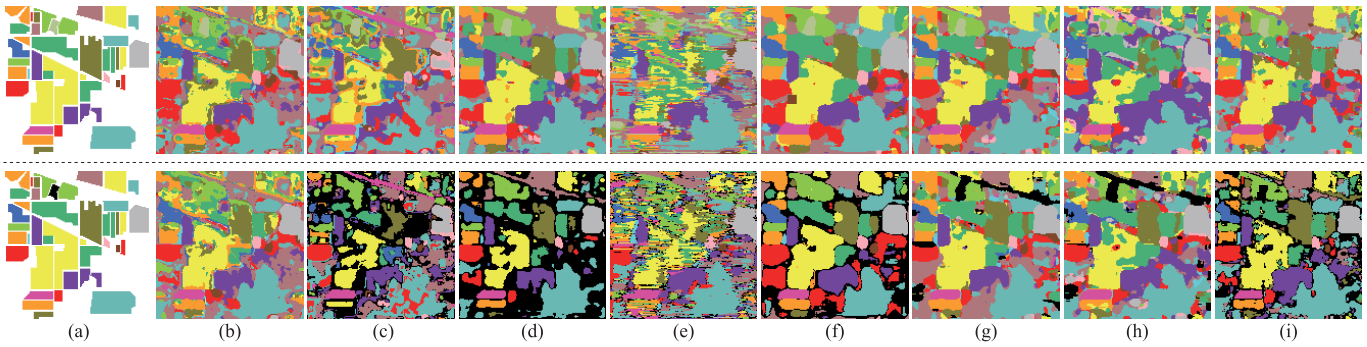


Fig. 11. Thematic classification maps of different methods on the IP scene. (a) Ground-truth map. (b) DFSL. (c) 3DOC-SSAN. (d) DBDA. (e) SSFTT. (f) SSTN. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL (the first row is the routine closed-set classification maps, and the second row is the corresponding open-set classification maps).

5) The conclusions above denote that the prevalent closed-set methods are still incapable of adapting to the risk of open space effectively. By contrast, three specialized open-set methods are relatively good at rejecting the unknown category and simultaneously retain a good performance on known categories. The fact denotes that open-set methods have a more practical value than idealized closed-set methods, because they can accommodate both the idealized closed-set hypothesis and the realistic open HSI world.
6) The proposed method SSMLP-RPL achieves the highest closed and open metrics among all the methods, and the highest accuracy of the unknown category (except for the IP data), which adequately verifies its superior performance both in CSE and OSE. Compared with the SOTA conventional method SSTN, SSMLP-RPL improves the closed OA of 0.16%, 1.96%, and 0.46% on IP, SA, and PU data, respectively. Compared with the latest SOTA open-set method SSLR, SSMLP-RPL improves the open OA of 2.30%, 3.22%, and 1.91% on IP, SA, and PU data, respectively. Therefore, SSMLP-RPL is more robust and effective for reducing open risk in the realistic open HSI world.

To further visually compare all methods, their corresponding open-set classification maps in the first trial are shown in Figs. 11–13. Besides, we also produce the conventional closed-set classification maps simultaneously to compare and analyze each method better, as shown in the first line of Fig. 11–13.

As we can observe, a conventional conclusion is that high closed metrics usually yield high-quality closed-set classification maps (it is not absolute), such as Fig. 11(d) and (f) (top), but the opposite leads to a bad effect, such as Fig. 11(c) and (e) (top). More importantly, all visual results can intuitively verify the above conclusions (1)–(6).

Specifically, DFSL cannot identify the unknown categories because of the closed property of SVM. DFSL still classifies every instance of the unknown category into different known categories. If other four closed-set methods do not deploy the softmax function with a probability threshold, they will also do the same as the DFSL.

The improved closed-set methods are capable of labeling the potential unknown category, which reflects in the black areas of open-set classification maps. But, the overall open-set effect is not satisfactory, which corresponds to conclusions (1) and (4). On the one hand, a prominent phenomenon is that they classify numerous known categories as the unknown category, such as Fig. 11(c) and (d) (bottom), leading to a large range of black areas that are not corresponding to the reality. On the other hand, even SSTN has a better open performance compared with other closed-set methods, it also cannot be effective in rejecting the unknown category, such as Fig. 11(f) (bottom).

As described in conclusion (3), an open-set classification map with good effect inevitably accompanies an excellent closed-set classification map, such as Fig. 11(f)–(i), while the reverse is not valid, such as Fig. 11(c).

As described in conclusion (2), the visual effect of open-set classification maps is usually worse, especially for the four conventional methods. But, the bad effect from the OSE plays a relatively minor impact on three specialized open-set methods.

As described in conclusion (5), classification maps of three specialized open-set methods achieve better visual effects both in the idealized closed-set hypothesis or the realistic open HSI world. They are usually corresponding to the authentic distributions of ground objects and have less number of noises, such as Figs. 11(g)–(i), 12(g)–(i), and 13(g)–(i).

As described in conclusion (6), SSMLP-RPL also has an excellent visual effect on closed-set and open-set classification maps, which verifies its robust and superior performance in the realistic open HSI world again. Furthermore, MDL4OW and SSLR usually lose some potential areas of unknown categories because of the fixed threshold of rejection, while SSMLP-RPL with the proposed dynamic learnable threshold is capable of exploring out more potential areas of unknown categories, which is more effective in adapting to the complex open world with novel categories of ground objects.

The discussion and analysis above are according to the assumption of the unknown category in Section III-A, which merely takes the last category of ground objects as the unknown category for each HSI data. To further verify the conclusions above in a larger risk of open space, we increase the number of unknown categories and simultaneously reduce the number of known categories for comparative experiments. According to the proportion of a certain category of ground objects, we choose a few categories that appear less frequently

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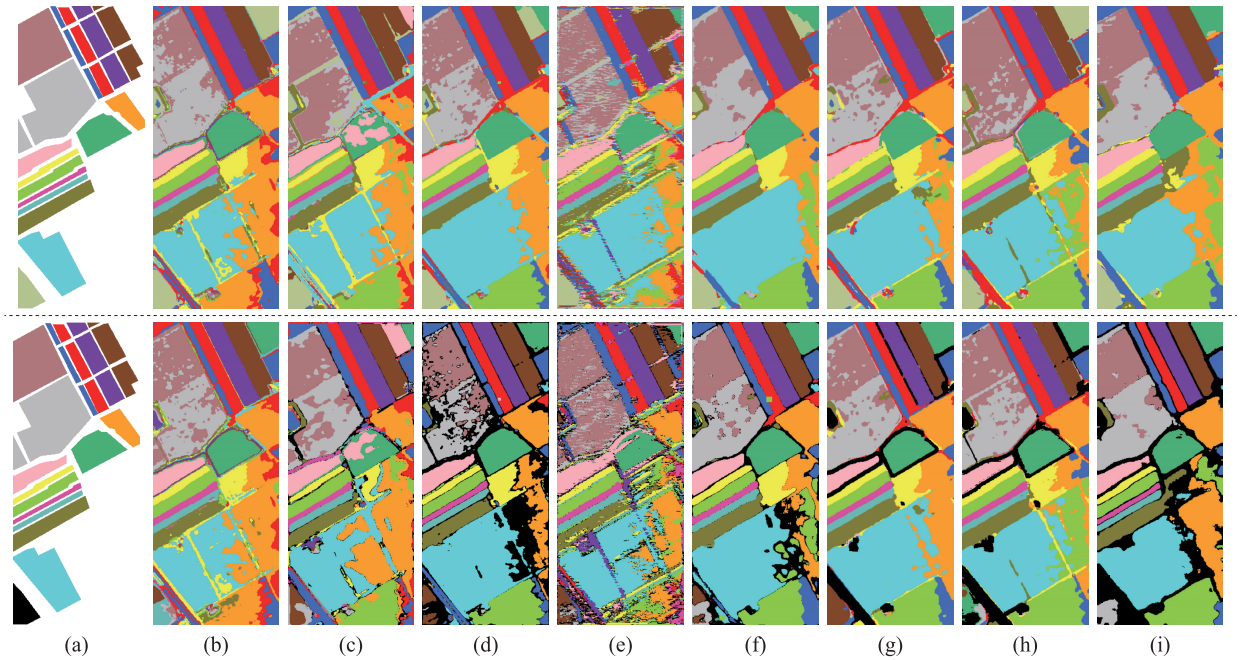


Fig. 12. Thematic classification maps of different methods on the SA scene. (a) Ground-truth map. (b) DFSL. (c) 3DOC-SSAN. (d) DBDA. (e) SSFTT. (f) SSTN. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL (the first row is the routine closed-set classification maps, and the second row is the corresponding open-set classification maps).

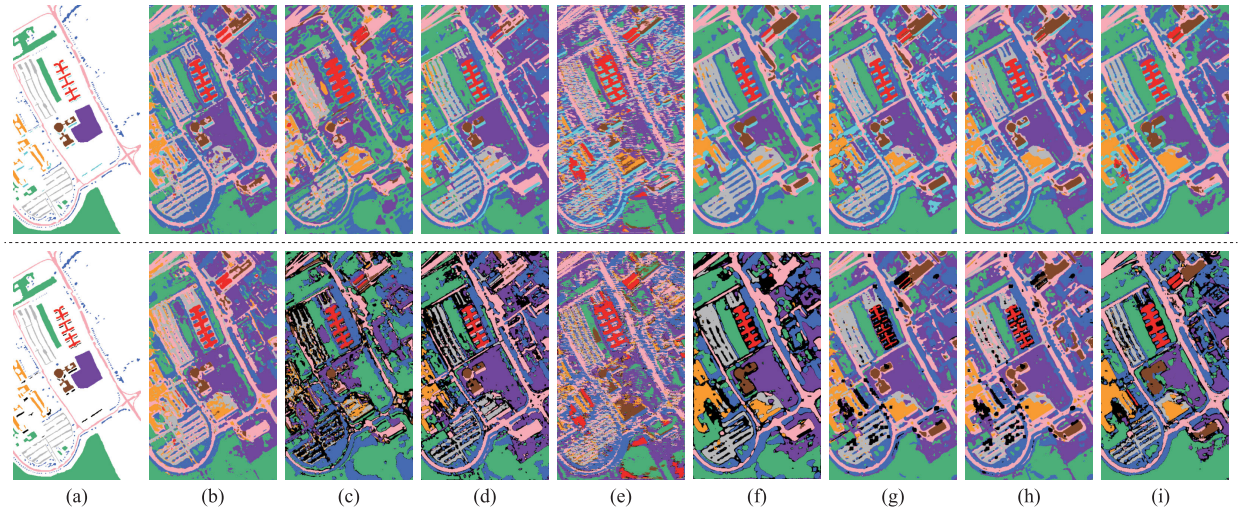


Fig. 13. Thematic classification maps of different methods on the PU scene. (a) Ground-truth map. (b) DFSL. (c) 3DOC-SSAN. (d) DBDA. (e) SSFTT. (f) SSTN. (g) MDL4OW. (h) SSLR. (i) SSMLP-RPL (the first row is the routine closed-set classification maps, and the second row is the corresponding open-set classification maps).

as the unknown categories. Specifically, the No. 1, No. 7, No. 9, and No. 16 categories are defined as the unknown category for IP data, the No. 11, No. 12, No. 13, and No. 14 categories are defined as the unknown category for SA data, and the No. 5, No. 7, and No. 9 categories are defined as the unknown category for PU data. Other detailed settings are consistent with the above experiment, and the results are shown in Tables V–VII.

As we can observe, conclusions (1)–(6) obtained above are still valid with the risk of open space expanding and the number of unknown categories increasing. Besides, we observe that the identification accuracy of the unknown categories usually decreases, as the number of unknown categories increases. While the closed metrics are less affected in this procedure, especially for the conventional closed-set methods, SSTN even achieves the highest closed metrics on IP and SA data. But, their degradation of open performance is more apparent. By contrast, the closed and open performances of specialized open-set methods are both less affected in this procedure, which denotes a better robustness when they face the real world with a larger risk of open space. SSMLP-RPL obtains better results among them, which further verifies its superior performance both in CSE and OSE.

F. Analysis of Trade-Off Weight

The trade-off weight λ in (11) decides the contribution of the margin loss function $\mathcal{L}_O$ during the fitting of model. The $\mathcal{L}_O$ utilized to constrain the open space and reduce the risk of open space is significant to the final open performance of

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TABLE V

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE IP SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 1, NO. 7, NO. 9, AND NO. 16 CATEGORIES ARE DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
2	66.96±2.65	67.94±8.09	86.07±3.58	40.95±9.24	88.84±3.20	84.94±3.46	82.36±5.68	85.64±3.40
3	85.86±3.42	63.09±8.42	87.86±5.50	51.49±7.64	89.72±2.68	86.92±3.92	83.43±6.27	86.96±3.12
4	97.21±1.89	90.67±4.78	98.77±1.07	84.97±6.47	98.27±1.56	94.39±2.11	93.31±3.10	95.07±4.68
5	94.40±3.13	89.19±2.06	95.38±2.07	70.53±4.37	93.33±2.27	91.43±3.21	93.98±3.99	90.70±3.75
6	96.68±2.36	91.90±4.42	98.54±1.63	82.98±5.06	98.20±1.00	98.19±2.64	92.14±6.47	94.30±3.62
8	99.49±1.13	99.37±0.99	99.97±0.06	94.53±3.14	99.97±0.06	99.98±0.06	96.80±6.74	97.81±2.91
10	75.55±7.51	79.94±7.19	88.80±5.28	55.26±10.76	88.34±4.98	91.02±3.88	87.35±7.73	90.46±6.72
11	66.49±3.69	64.61±9.32	79.49±4.24	50.04±8.77	78.96±6.26	81.23±4.76	79.68±6.49	85.95±1.76
12	66.93±4.85	73.27±8.79	95.93±2.81	56.03±7.98	94.89±2.82	88.90±3.41	82.65±8.71	97.56±1.03
13	99.56±0.70	99.46±0.51	99.75±0.33	96.43±2.56	99.60±0.42	99.95±0.15	96.71±4.25	96.80±2.62
14	92.80±3.24	91.34±3.46	95.96±2.06	84.30±6.19	95.53±2.99	94.13±2.06	94.40±3.55	92.20±2.81
15	93.08±3.13	81.01±7.20	96.68±3.51	77.74±8.33	96.60±1.84	90.98±4.90	89.09±5.12	95.44±4.17
Unknown	0.0±0.0	3.47±1.59	16.79±4.09	11.44±5.20	20.10±7.34	48.24±1.91	50.32±0.61	**59.87**±6.41
Closed OA	79.88±1.24	80.63±2.80	92.15±0.94	63.59±1.69	**92.97**±1.44	90.56±1.28	86.81±2.20	92.21±0.95
Closed AA	86.25±1.03	85.27±2.51	93.85±0.57	71.55±1.25	**95.39**±0.71	94.01±0.85	90.93±1.80	94.98±1.24
Closed Kappa	77.21±1.39	78.05±3.15	91.20±1.07	59.05±1.77	**92.13**±1.63	88.33±1.75	84.59±2.12	91.86±1.49
Open OA	78.42±1.22	75.73±3.91	87.37±0.97	61.42±1.71	87.54±1.71	87.18±1.22	85.34±2.03	**88.57**±2.37
Open AA	79.62±0.95	76.56±3.09	85.69±0.65	65.90±1.09	86.87±1.18	87.48±0.71	86.71±1.02	**88.83**±1.81
Open Kappa	75.65±1.36	72.90±4.23	85.92±1.07	56.96±1.76	86.13±1.88	86.63±1.35	83.40±2.07	**86.89**±2.58

TABLE VI

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE SA SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 11, NO. 12, NO. 13, AND NO. 14 CATEGORIES ARE DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
1	98.62±0.89	97.22±1.55	99.97±0.09	83.52±24.49	99.82±0.22	92.49±0.65	94.93±3.65	99.44±0.61
2	99.46±0.43	99.30±1.59	100.0±0.0	78.66±12.82	99.99±0.00	95.99±0.55	98.66±0.87	99.18±0.87
3	98.83±2.00	97.18±2.93	99.98±0.05	87.47±6.30	99.91±0.20	99.86±0.23	99.32±0.56	99.10±1.12
4	99.51±0.39	99.49±0.49	99.67±0.26	93.88±2.33	99.39±0.42	99.13±0.63	99.00±0.86	98.49±1.40
5	97.76±2.00	95.53±2.11	98.41±0.79	88.40±3.64	97.09±2.31	91.82±1.79	97.79±1.88	98.41±1.56
6	99.20±1.62	99.06±0.81	99.84±0.19	95.02±3.38	99.31±0.96	96.88±1.87	97.12±3.28	96.91±2.68
7	99.86±0.16	99.56±0.22	99.91±0.06	92.76±3.36	99.59±0.42	95.94±0.51	95.41±1.18	98.85±1.27
8	79.51±5.10	63.83±13.00	79.75±5.83	49.78±13.18	80.41±9.28	81.96±4.55	83.06±5.73	80.55±7.07
9	98.55±1.41	99.18±1.28	99.81±0.20	92.38±3.90	99.96±0.05	99.92±0.03	99.00±0.34	99.52±0.46
10	92.91±2.97	95.22±0.89	96.13±0.72	85.69±10.93	96.49±1.43	97.75±1.02	96.72±1.13	97.36±1.28
15	86.29±3.72	81.19±5.78	83.97±4.66	63.53±13.24	87.20±6.45	85.97±8.17	92.52±7.74	82.72±6.26
16	98.41±1.18	96.96±1.20	98.62±0.52	87.15±3.81	98.48±0.57	98.82±0.89	97.89±1.09	98.99±0.31
Unknown	0.0±0.0	5.03±3.83	23.06±5.60	6.76±3.86	26.21±14.84	31.42±11.10	40.27±9.04	**52.85**±7.09
Closed OA	92.20±0.66	87.97±2.52	93.44±0.82	76.89±2.90	**94.22**±1.38	93.47±0.58	92.63±2.31	93.84±1.36
Closed AA	95.74±0.30	93.80±0.97	96.84±0.27	83.83±2.89	**97.19**±0.54	96.82±0.45	96.39±0.89	96.95±0.98
Closed Kappa	91.16±0.73	86.42±2.79	92.55±0.93	74.01±3.20	**93.44**±1.55	92.68±0.81	91.45±1.76	93.07±1.24
Open OA	83.72±0.60	80.13±2.42	86.14±0.83	69.79±2.78	86.90±2.45	85.53±0.61	85.71±2.16	**88.54**±2.55
Open AA	88.38±0.28	86.83±1.03	90.70±0.37	77.31±2.77	91.07±1.53	87.46±0.57	88.21±0.93	**92.49**±1.33
Open Kappa	81.79±0.65	77.93±2.62	84.53±0.91	66.54±3.01	85.39±2.69	82.63±0.69	83.64±2.36	**87.21**±2.83

model. To analyze the impact of trade-off weight λ, we set λ to 0, 0.01, 0.1, 0.2, 0.5, and 1, respectively, to observe the closed and open performances of SSMLP-RPL on three HSI data. Also, the results are shown in Fig. 14. As we can observe, different trade-off weights λ have an ignorable effect on the closed performance of SSMLP-RPL, while have a significant effect on the open performance. The open performance of SSMLP-RPL is unsatisfying without the constraint to the open space ($\lambda = 0$). To a certain extent, the increase of trade-off weight λ is beneficial to improve the open performance of SSMLP-RPL. However, such improvement becomes less noticeable when λ>0.1. Therefore, an appropriate $\lambda = 0.1$ can balance the contribution of the margin loss function $\mathcal{L}_O$ to improve the final open performance of SSMLP-RPL.

G. Analysis of SSMLP Network

In fact, the proposed SSMLP-like network can also be taken as a conventional end-to-end classifier, and the prominent characteristic of SSMLP distinguished from the current works is that SSMLP merely utilizes a series of simple FC layers to capture spectral–spatial feature for classification. To further verify the performance of SSMLP, we compare it with currently SOTA CNN-based and Transformer-based methods. Because the primary purpose here is to verify the performance of different backbones, the conventional closed-set assumption is followed. To comprehensively compare with representative works of different backbones, except for five conventional methods above, we add several representative methods, involving two influential SVM-classifier methods: 3-D convolutional autoencoder (3DCAE) [41] and 3-D autoencoder with Siamese network (3DAES) [46], one influential CNN-based method: spectral–spatial residue network (SSRN) [21], one influential Transformer-based method: SpectralFormer [35], and one pioneering MLP-based method: MUMLP [74]. As for the SSMLP here, we set S to 13 and L to 1 as the standard configuration according to the analysis in Section III-F. The average OA (%) of different methods on three HSI data is shown in Table VIII.

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TABLE VII

ANALYSIS RESULTS OF ACCURACY WITH DIFFERENT METHODS ON THE PU SCENE (%) (30 TRAINING SAMPLES IN EACH CATEGORY; *THE OUTPUT WITH THE MAXIMAL PROBABILITY < 0.5 CONSIDERED AS THE UNKNOWN CATEGORY; THE ACCURACY PER CATEGORY IS CALCULATED UNDER THE OPEN ENVIRONMENT; THE NO. 5, NO. 7, AND NO. 9 CATEGORIES ARE DEFINED AS THE UNKNOWN CATEGORY)

Category	DFSL [49]	3DOC-SSAN* [26]	DBDA* [24]	SSFTT* [37]	SSTN* [34]	MDL4OW [65]	SSLR [66]	SSMLP-RPL
1	90.33±3.84	83.25±3.78	97.75±0.94	74.45±6.38	96.94±1.20	92.81±1.61	91.23±2.99	97.79±1.99
2	92.90±1.70	76.22±6.41	91.43±4.38	78.78±6.09	92.33±4.39	96.55±2.03	95.86±1.97	96.10±1.97
3	84.92±3.28	66.00±8.95	92.39±4.24	60.68±7.67	94.83±2.48	91.06±6.06	88.51±7.70	92.22±1.89
4	96.56±1.08	91.84±3.47	96.29±1.51	91.78±5.50	93.92±2.33	88.53±3.06	94.94±1.34	91.68±0.91
6	94.59±2.83	74.47±8.35	97.77±2.08	62.15±9.44	99.09±0.81	95.89±1.26	95.24±2.23	98.36±2.19
8	84.41±4.88	73.53±9.01	92.52±3.35	63.68±9.27	91.30±3.71	88.19±2.56	88.65±3.48	91.24±2.55
Unknown	0.0±0.0	2.36±2.63	1.02±1.95	3.64±1.01	4.79±3.79	43.21±2.01	48.53±6.80	**54.25**±6.52
Closed OA	91.74±0.94	78.43±3.32	94.85±1.73	74.97±2.56	95.39±1.95	95.78±0.89	95.85±0.92	**97.01**±0.83
Closed AA	90.62±0.94	78.57±2.85	95.74±0.82	72.47±2.36	96.20±0.91	96.46±0.99	96.87±1.03	**97.17**±1.92
Closed Kappa	88.54±1.28	70.84±4.12	92.90±2.33	66.08±3.01	93.63±2.62	93.79±0.99	94.09±1.47	**94.99**±1.77
Open OA	83.97±0.86	71.23±2.89	85.99±1.71	68.53±2.36	86.58±1.94	89.51±0.69	90.01±0.88	**90.95**±1.72
Open AA	77.67±0.80	66.81±2.69	81.31±0.76	62.16±2.10	81.89±1.18	86.04±1.12	86.14±0.83	**87.52**±1.78
Open Kappa	78.73±1.11	62.75±3.40	81.52±2.08	59.16±2.62	82.33±2.40	86.32±0.88	86.73±1.14	**87.43**±2.19

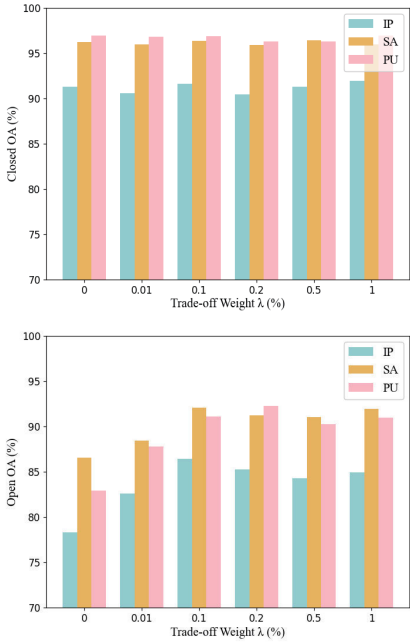


Fig. 14. Bar chart of the closed and open performances varying with trade-off weight λ.

TABLE VIII

OA OF DIFFERENT METHODS ON THE THREE HSI DATA (%) (30 TRAINING SAMPLES IN EACH CATEGORY)

Method		IP	SA	PU
SVM-classifier	3DCAE [41]	71.18	88.05	80.08
	DFSL [49]	80.25	91.19	90.42
	3DAES [46]	83.21	89.04	85.62
CNN-based	3DOC-SSAN [26]	78.98	82.89	75.22
	SSRN [21]	90.36	94.29	91.73
	DBDA [24]	**92.32**	94.67	92.38
Transformer-based	SpectralFormer [35]	58.81	68.97	83.05
	SSFTT [37]	64.29	72.46	74.10
	SSTN [34]	91.24	**95.14**	92.99
MLP-based	MUMLP [74]	69.39	86.79	81.49
	SSMLP	91.33	94.85	**95.13**

As we can observe, performance of methods based on SVM classifier heavily relies on the discriminative power of extracted feature. Thus, they can obtain satisfactory results with a two-stage mode (feature extraction and classification). But, they generally cannot compete with advanced end-to-end classification methods. Currently advanced end-to-end HSI classification methods can be roughly divided into two types based on the backbone, involving the CNN-based and the Transformer-based. These works achieve huge development recently, but the authentic performance under the condition of limited samples size is uneven. As an initial work of MLP, although MUMLP merely considers the spectral feature, it surpasses the influential SpectralFormer on three data, showing the advantage of MLP architecture for encoding spectrum. But, MUMLP still cannot compete with SOTA works. Obviously, there are a large scope for performance improvement by using MLP architecture to capture spatial semantics. The proposed SSMLP network is verified to be capable of competing with the SOTA works based on CNN and Transformer on different HSI data, simultaneously obtaining the highest accuracy on the PU data. Therefore, SSMLP demonstrates the feasibility and huge potential of MLP backbone for spectral–spatial feature extraction.

H. Ablation Experiment

To further analyze different components of SSMLP-RPL, we design the ablation experiment to verify their effectiveness. Specifically, the original RPL embedded with the nine layers of convolution network is taken as the baseline method [68], and the relevant hyperparameters, such as patch size S, are set to be consistent with this article to control the variables. Other specifical schemes A–E are utilized to verify the effectiveness of each proposed component by combining different designs. The results of ablation experiment are shown in Table IX.

First, the performance improvement of scheme **D** compared with the baseline shows the superiority of SSMLP network when processing HSI data. Specifically, the closed OA of three HSI data is increased by 12.74%, 3.31%, and 8.91%, respectively, and the open OA of three HSI data is increased by 10.65%, 3.01%, and 4.72%, respectively. Simultaneously, the performance improvement of scheme **E** compared with scheme **C** also confirms the fact.

Second, the performance improvement of scheme **C** compared with the baseline shows the superiority of the strategy of learnable dynamic threshold when facing the unknown categories. Therefore, the strategy of learnable dynamic threshold can effectively enhance open performance of the model

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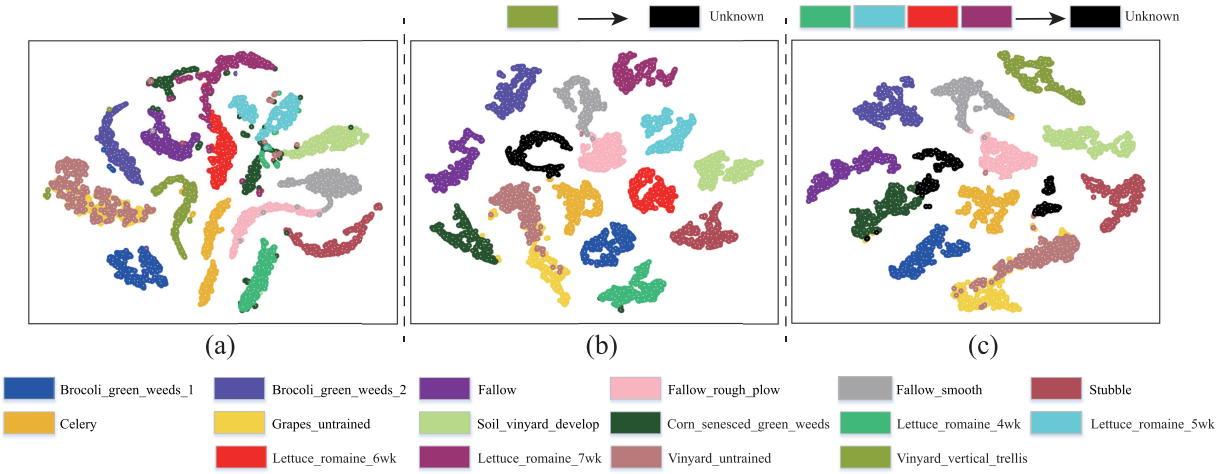


Fig. 15. T-SNE visualization results of feature space on SA data. (a) Original feature space. (b) Feature space constructed by SSMLP-RPL (the No. 16 category is defined as the unknown category). (c) Feature space constructed by SSMLP-RPL (the No. 11, No. 12, No. 13, and No. 14 categories are defined as the unknown category).

TABLE IX

RESULTS OF ABLATION EXPERIMENT ON THE THREE HSI DATA (%) (30 TRAINING SAMPLES IN EACH CATEGORY)

		Baseline	A	B	C	D	E
SpeFC Module			✓			✓	✓
SpaFC Module				✓		✓	✓
Dynamic Threshold			✓	✓	✓		✓
IP	Unknown	23.98	73.81	23.06	40.73	28.59	**74.74**
	Closed OA	77.40	90.49	90.27	77.09	90.14	**91.60**
	Open OA	74.91	75.82	84.76	76.17	85.56	**86.45**
SA	Unknown	27.53	53.08	14.07	38.62	36.47	**62.39**
	Closed OA	90.17	95.41	84.89	90.04	93.48	**96.39**
	Open OA	87.18	88.40	81.40	88.38	90.19	**91.17**
PU	Unknown	32.92	64.17	23.69	47.84	35.81	**70.71**
	Closed OA	87.55	95.66	90.11	87.72	96.46	**96.87**
	Open OA	85.66	82.41	85.13	86.94	90.38	**91.11**

and maximize the rejection of unknown categories without affecting the identification of known categories as far as possible. Specifically, the open OA of three HSI data is increased by 1.26%, 1.20%, and 1.28%, respectively, and the accuracy of the unknown category on three HSI data is increased by 16.75%, 11.09%, and 14.92%, respectively. Also, the performance improvement of scheme **E** compared with scheme **D** also confirms the fact.

Third, the comparison results between schemes **A** and **B** show that the SpeFC module makes a greater contribution than the SpaFC module. Also, the performance of the model merely with SpeFC module can achieve a closed performance close to scheme **E**, which denotes its significance. But, the single SpeFC module cannot adapt to the open-set HSI classification well, because the single feature is not robust enough to reject the unknown categories.

At last, compared with schemes **A** and **B**, the performance improvement of scheme **E** reflects the superiority of the combination of the two modules. In particular, compared with scheme **A**, the open OA of three HSI data is increased by 11.54%, 2.77%, and 8.70%, respectively, and the accuracy of the unknown category on three HSI data is increased by 0.93%, 9.31%, and 6.54%, respectively. It also denotes the necessity of SpaFC module, because the spectral–spatial feature is more robust when facing the unknown categories. Scheme **E** is the final version of the proposed method, which achieves the highest results on each metric for three HSI data. It also verifies the superior performance of the final combination of different components both in CSE and OSE.

I. Analysis of Learnable Dynamic Threshold

To further intuitively display the effectiveness of learnable dynamic threshold, we show the final learned results according to (13), as shown in Table X. As we can observe, the proposed strategy of learnable dynamic threshold can indeed obtain thresholds adapted to different categories according to the minimum distance between different categories and their reciprocal points. Due to the complex distribution of different ground objects in feature space, the distances between different categories and their reciprocal points $d_e{}^k$ are usually quite different from each other. Compared with a single and fixed threshold, the dynamic thresholds can better identify some unknown categories that are closer to the known category with a larger d_e, while the single and fixed threshold may fail to reject these potential and difficult unknown categories. Therefore, the dynamic thresholds can maximize the rejection of unknown categories without affecting the identification of known categories as far as possible. In general, the proposed strategy of learnable dynamic threshold can improve open performance of the model, and the laborious determination of specific threshold can be avoided.

J. Analysis of RPL

To further intuitively display the discriminative feature space constructed by SSMLP-RPL, we utilize the T-SNE dimension-reduction visualization approach to further present distinctly its advantage of identifying known and unknown categories. We take SA data as an example to display, and the visualization results are shown in Fig. 15.

As we can observe, the distinguishability of original SA data in feature space is worse, which reflects in the overlapped and the close adjoining between different categories. After training the SSMLP network with RPL, SSMLP can project data instances into a satisfactory high-dimensional feature space for open-set HSI classification. Specifically, the distance between

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TABLE X
RESULTS OF LEARNED DYNAMIC THRESHOLD ON THREE HSI SCENES (UNDER THE ASSUMPTION OF THE UNKNOWN CATEGORY IN SECTION III-A)

k	IP			SA			PU		
	${d_e}^k$	$\mathcal{L}_O$	$\mathcal{T}^k$	${d_e}^k$	$\mathcal{L}_O$	$\mathcal{T}^k$	${d_e}^k$	$\mathcal{L}_O$	$\mathcal{T}^k$
1	4.91	1.55	3.36	7.90	2.16	5.74	3.63	1.73	1.90
2	3.42	1.55	1.87	8.90	2.16	6.74	6.58	1.73	4.85
3	2.83	1.55	1.28	9.53	2.16	7.37	4.74	1.73	3.01
4	4.27	1.55	2.72	4.82	2.16	2.66	6.40	1.73	4.67
5	5.26	1.55	3.71	10.20	2.16	8.04	7.22	1.73	5.49
6	5.11	1.55	3.56	8.92	2.16	6.76	6.04	1.73	4.31
7	3.84	1.55	2.29	9.45	2.16	7.29	6.69	1.73	4.96
8	5.86	1.55	4.31	7.11	2.16	4.95	5.26	1.73	3.53
9	5.59	1.55	4.04	10.09	2.16	7.93			
10	2.88	1.55	1.33	8.92	2.16	6.76			
11	4.26	1.55	2.71	8.51	2.16	6.35			
12	6.21	1.55	4.66	7.15	2.16	4.99			
13	5.40	1.55	3.85	9.29	2.16	7.13			
14	5.15	1.55	3.60	4.95	2.16	2.79			
15	4.19	1.55	2.64	7.98	2.16	5.82			

the known categories and the corresponding reciprocal points is enlarged, so the known categories are usually pushed to the periphery of the space by the corresponding reciprocal points, while the distance between the potential unknown categories and the reciprocal points is constrained to a limited range. Therefore, in the feature space constructed by SSMLP-RPL, the distance between known categories and unknown categories is enlarged, so that the unknown categories can be identified easier. A fine feature space ensures that the known categories can be classified reasonably, while the unknown categories are rejected. More importantly, the label of an instance to be classified will be determined by the distance with the reciprocal points, which can reject the unknown categories by a reasonable threshold and effectively reduces the risk of open space. Besides, a single and fixed threshold may lead to lose potential unknow categories, because the distance between a known category and its reciprocal point is not always consistent for all known categories. Therefore, the proposed strategy of learnable dynamic threshold can maximize the rejection of unknown categories without affecting the identification of known categories as far as possible.

IV. CONCLUSION

Although deep-learning-based HSI classification methods have achieved huge development in recent years, they generally operate with an idealized assumption of the CSE. However, the realistic open world is an objective existence, which cannot be neglected in the development of HSI classification. The prevalent closed-set methods still cannot effectively and robustly handle unknown categories of ground objects in OSE. Therefore, we propose an SSMLP-RPL to improve the performance of open-set HSI classification.

The motivation of our work involves two aspects. First, as a novel learning paradigm, RPL enables to better model extra-category space and effectively reduce the risk of open space, so we introduce it to solve the problem of open-set HSI classification for the first time. Also, we further propose a strategy of learnable dynamic threshold to improve the open performance of the model effectively. Second, aiming at better modeling discriminative dependency of spectrum and capturing conducive spatial semantics, motivated by the huge potential of MLP backbone, we construct an SSMLP-like feature extraction framework composed by a series of simple FC layers. SSMLP can obtain a robust feature representation to construct exquisite feature space and identify known and unknown categories.

To verify the performance of SSMLP-RPL, we conduct abundant experiments and obtain valuable conclusions. We practically demonstrate that the currently closed-set end-to-end frameworks can be easily transformed into the open-set classification methods. Also, their open performance is strongly associated with the initial closed performance. Specialized open-set methods generally have a good performance in CSE, so their superior capacity of feature extraction guarantees the rejection of the unknown categories. Therefore, open-set methods can better accommodate both the idealized closed-set hypothesis and the realistic open world. SSMLP-RPL has a competitive performance both in CSE and OSE and even surpasses currently advanced closed-set and open-set HSI classification methods, which is more robust and effective for reducing open risk in the realistic open HSI world. Besides, as a representative work of MLP-backbone, the SSMLP network can compete with the SOTA works based on CNN and Transformer, which shows the feasibility and huge potential of MLP-backbone for HSI classification.

However, we note that SSMLP-RPL usually misidentifies the boundaries between different ground objects as the unknown categories, which is because the samples of ground object boundaries are usually far from the category center of the distribution in feature space. Therefore, how to better constrain the distance between the samples of boundaries and the category center of the distribution and effectively mitigate such misidentification phenomenon is worthy of further attention and exploration.

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